Team Control Number

15072

Problem Chosen

B

2024
HiMCM
Summary Sheet

Sustainable Bytes: Unveiling the Energy, Emissions, and Health Costs of High-Powered Computing

In an era of increasing demand for High-Powered Computing (HPC) in sectors such as artificial intelligence (AI), data science, and cryptocurrency mining, the energy-intensive computational process has raises significant concerns about their environmental consequences. Our study provides a comprehensive model to assist policymakers in understanding the potential burden placed on electricity system and their associated environmental and health impacts .

Our study is structured with three models. The first model quantifies HPC energy consumption and associated carbon emission for **each country** with a thorough **life cycle analysis**. Results are presented at three levels: globally, by continent, and for the top 10 countries. The analysis highlights that North America, Asia, and Europe dominate HPC capacity, with the United States and China emerging as the clear leaders. Globally, HPC operations consume 36.4 TWh of electricity annually, resulting in total emissions of 18.8 MtCO2e this year.

Our second focus is on developing a forecasting model for HPC capacity and electricity demand. This includes predicting global HPC demand growth using a **Logistic Growth Model** and forecasting individual country trends with a **Linear Regression Model**. The model also incorporates future improvements in data center Power Usage Effectiveness (PUE) using an **Exponential Decay Model**. Projections indicate that by 2030, global HPC energy consumption could reach 47.0 TWh, resulting in 24.4 MtCO2e of emissions. To assess the long-term implications of rapidly growing HPC energy demand, the model employs a **Dynamic Competition Model** to predict interactions between HPC and other electricity-demanding sectors. The findings suggest that fast-growing sectors like HPC and transportation could begin to outcompete slower-growing sectors for electricity. A sensitivity analysis tests model variability against HPC growth rates and PUE values, while further studies explore potential carbon emission reductions achievable through higher green energy integration and improved PUE.

We then extend our model to evaluate the **air quality** and **health impacts** associated with HPC operations. Specifically, we assess PM2.5 emissions from HPC data centers in the top 20 countries, quantifying both current and future health damages. Our analysis incorporates key factors such as population growth, renewable energy penetration, dose-response relationships with relative risks, and monetary valuation using the Value of Statistical Life (VSL) methodology. By 2030, our model predicts that air pollution from HPC-related activities could result in over 6 thousands premature deaths and economic losses exceeding $5 billions USD.

Our models provide a **robust framework** for evaluating the HPCs' energy consumption, life-cycle carbon emissions and environmental impacts. This work paves the ways for better-informed decisions on striking a balance between satisfying the high-speed growth of HPC demand and reducing its negative environmental impacts with further consideration of increasing renewable penetration.

Keywords: HPC; Energy Consumption, Life-Cycle Analysis, Carbon Emission; Health Damages

Team Control Number

15504

Problem Chosen

A

2024

HiMCM

Summary Sheet

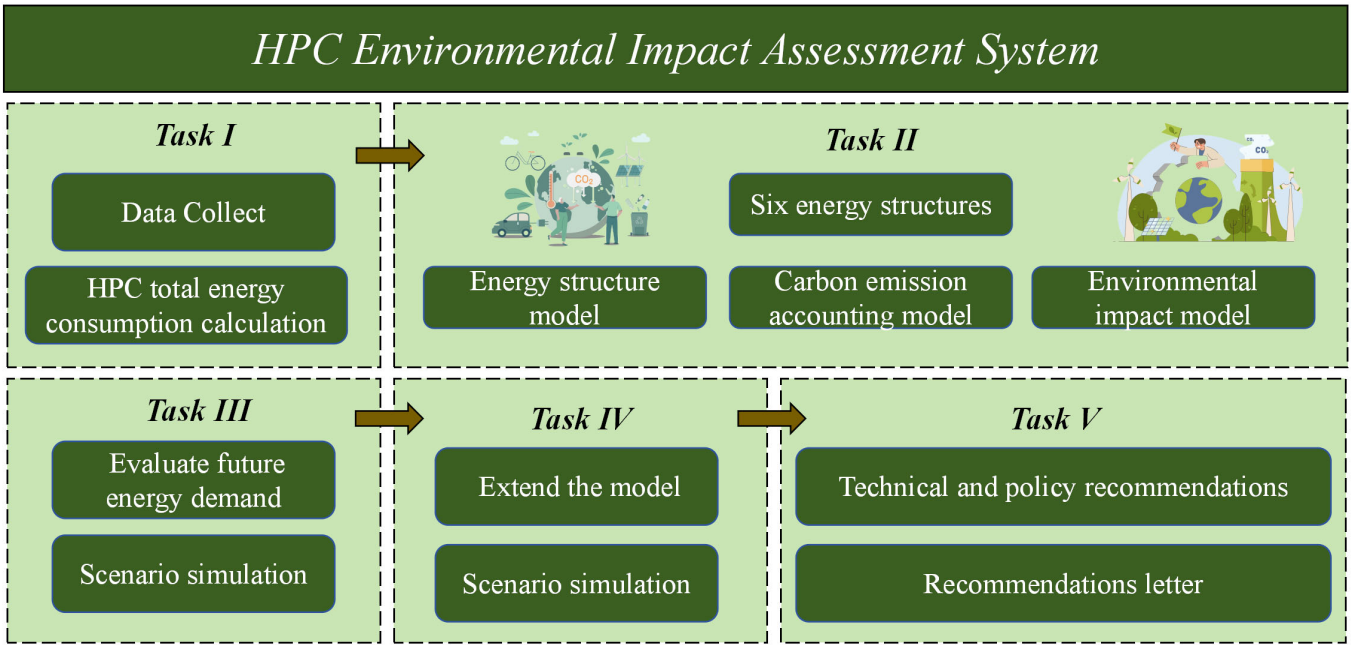
The Olympic Games, a quadrennial global sporting spectacle, continually evolves with certain sports/disciplines/events (SDEs) being added or removed. The International Olympic Committee (IOC) employs six criteria to evaluate potential additions. The problem requires us to develop a model based on factors in line with IOC's criteria, predicting the inclusion of SDEs in future Olympics.

The core challenge of this problem is a binary classification task: determining whether a specific SDE should be included or excluded from the Olympics. As constrained by a limited number of SDEs per Olympic Games, the selection process prioritizes SDEs based on their relative significance. Probability offers a robust tool to assess the likelihood of an SDE's inclusion in alignment with Olympic values.

We developed a logistic regression model that incorporates three key items (Problem 2): a base probability (P_0) that evaluates the adherence to IOC's values based on nine quantitative and qualitative subcriteria we developed, an inertia component (P_{in}) accounting for the effect of past inclusion of an SDE in the Olympics, and a bias-correction item (P_{it}) that offers the flexibility to adjust the model best match the historical data. We developed nine subcriteria and collected historical data on Olympics sports from the Internet and literatures (Problem 1). Our model demonstrates strong predictive accuracy, accurately reproducing historical inclusion patterns, such as continuous presence, newly added or removed (Problem 3).

Sensitivity analysis revealed that the Popularity and Relevance and Innovation criteria are most influential factors in determining an SDE's likelihood of inclusion (Problem 5). According to our model prediction, in the 2032 Brisbane Olympics, Australian football takes the most potential to be included, followed by Zwift cycling and netball, among a total of 7 candidate sports including most popular ones in Australia (Problem 4). As Zwift virtual cycling becomes more popular among young people, it is more likely to be included in 2036 or beyond (Problem 4).

Compared to traditional methods such as the Analytical Hierarchy Process, our approach offers a more objective and flexible framework. Unlike subjective expert judgements, our model learns and constrains the weights and parameters from historical data to identify patterns and trends. While data collection remains a significant challenge, future efforts in collecting more comprehensive data can further enhance the model's predictive power. Our current model operates at the discipline level, it can be extended to the event level provided event-specific data is available. This would allow for even more granular predictions and insights into the future of the Olympic Games.


HPC Environmental Impact Assessment System
Task I
Data Collect
HPC total energy consumption calculation
Task II
Six energy structures
Energy structure model
Carbon emission accounting model
Environmental impact model
Task III
Evaluate future energy demand
Scenario simulation
Task IV
Extend the model
Scenario simulation
Task V
Technical and policy recommendations
Recommendations letter

1.3 Assumptions and Justifications

1. **Assumption** : The number of continents with at least one member country in the specific sports governing body recognized by the IOC need not matter in regards to the diversity, equity and inclusivity of the geopolitical reality of such sport, as long as such number exceeds or is equal to the number of four.
   Justification : The number of continents has a narrow range of $4 \sim 5$. Compared to the number of countries that has a range of $75 \sim 227$, the number of continents provides minimal statistical significance, thus an inadequate parameter for analysis.

[1] three Olympic Games were canceled due to the world wars

2. **Assumption** : Each sport is responsible for all the economical costs and environmental impacts in all venues where such sport is to be hosted, even if the aforementioned venues are to be shared with other sports events.
   Justification : For our future generations, reuse of venues is strongly encouraged. However, this leads to a very philosophical question about causality. If sport A and sport B use the same venues with a cost of C, does A *cause* C? For the purposes of this model, sport A is totally responsible for the Cost of the venue, as A is a sufficient condition of C.

3. **Assumption** : Olympic standard venues of the same type, provided that they have been recently constructed,[2] have negligible difference in the inflation-adjusted cost, environmental impact, and other parameters.
   Justification : Olympic venues can be designed in different styles, often utilizing different materials with different costs and environmental footprints, but analyzing the architectural values and costs is beyond the nature of sports itself. To provide a fair analysis of all sports alike, architectural consideration will not be evaluated.

4. **Assumption** All Olympic standard venues constructed recently are in accordance with the regulations and standards of sports hosted, and all outdoor sports are hosted in similar conditions, implying a similar injury rate.
   Justification During *Paris 2024*, concerns about the water quality in which athletes competed in did exist.[2] However, analyzing the safety of the venues is beyond the scope of the topic, which is to evaluate the sports themselves. Therefore, it is assumed that all venues are within safety standards, and as such the injury rate of a given sport is only related to the nature of such sport.

5. **Assumption** All injuries, as long as it leads to one or more day off, are considered severe.
   Justification Since all injuries can and do lead to a possibility of an athlete's reduction of performance and basic function, sometimes permanently, it is important to consider all injuries when evaluating safety.

6. **Assumption** Carbon emission of constructing an arena is dependent on the amount of concrete.
   Justification The majority of carbon emission comes from the manufacturing of concrete. The inherit chemistry of calcium carbonate, the core ingredient of concrete, is responsible for 60% of the carbon emission,[3] which is not improvable by any means. Therefore, it can be safely assumed that the amount of carbon emission is mainly dependent of the amount of concrete used in construction.

7. **Assumption** The amount of concrete used in the construction of an arena is dependent on its type and its area.
   Justification As in assumption 3, the purpose of the model is not to judge the architectural design of the arena. Considering that there are meaningful differences in

[2] within twenty years, inclusive

the nature of arenas for different sports, only the arena's type and capacity will be considered.

8. **Assumption** Renovation for the purpose of upgrading to Olympic standard does not have a significant carbon footprint.
   Justification While it is true that renovation would inevitably lead to some carbon emission, it is minuscule if compared to constructing a new one. Thus it would not be taken into account.

9. **Assumption** All sports venues will be used and maintained everyday in the time span of the Olympic.
   Justification It is expected that all resources will be used, especially considering there are few venues qualified for the Olympic Games.

10. **Assumption** Google Trends™ can accurately represents global searches.
    Justification Since Google is the most used search engine by a large margin[4], and that search engine is generally not correlated to a preference in sports, it can be safely assumed that Google Trends depicts a certain sport's popularity accurately.

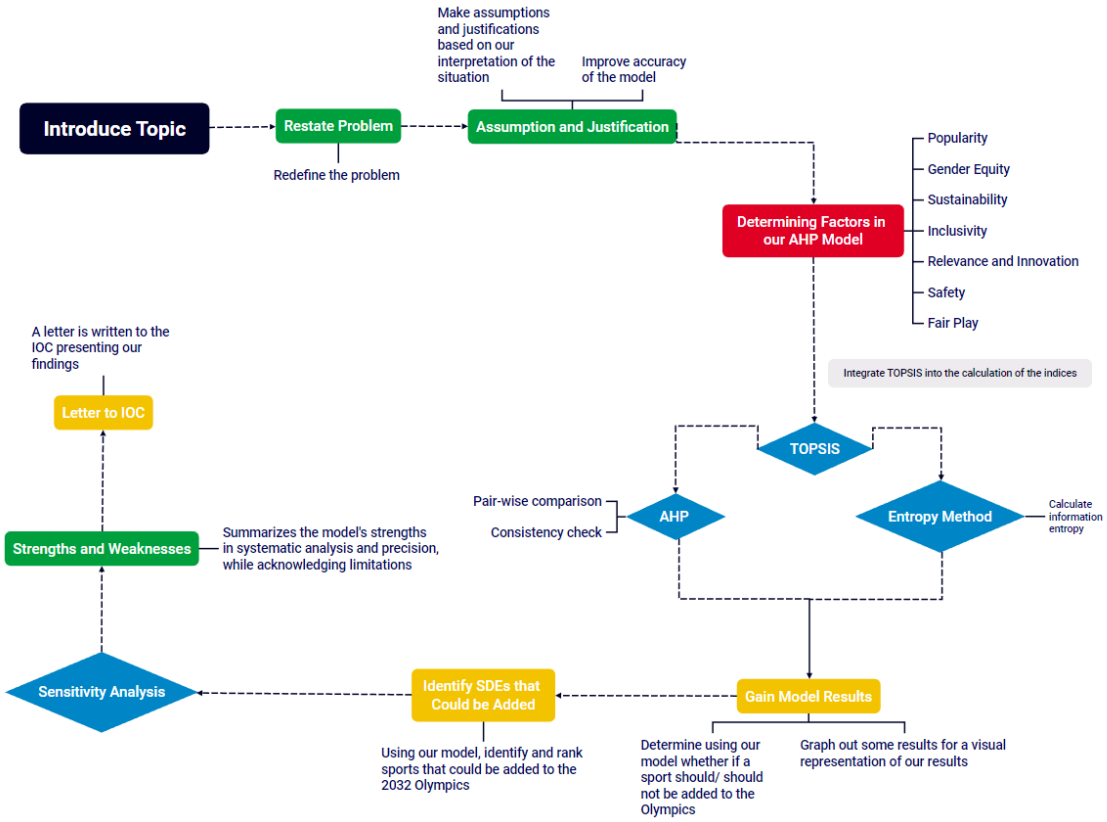


Figure 1: Work Structure

2 Assumptions and Justifications

1. **Assumption 1: Energy consumption is linearly proportional to HPC computational load**

High-performance computing systems typically adjust resource usage based on computational demand. Assuming a linear relationship between energy consumption and computational load simplifies the analysis and provides a foundation for more complex, non-linear models. This assumption allows the total energy demand of HPC to be estimated using data on full capacity and average utilization.

2 Table of Variables

Variables	Descriptions (Unit)	[1]	[2]	[3]
I_{GInc}	Geographical Inclusivity Index			D
v_{State}	Number of Eligible Countries in a Sport	□	V	D
v_{Ctn}	# of Continents w. min. 1 Eligible Country	□	V	D
I_{Gen}	Proportional Gender Representation Index			D
Δv_{Gen}	Diff. between Proportion of Genders	□	V	D
v_{GenM}	Proportional Representation of Male Athletes	□	V	D
v_{GenF}	Proportional Representation of Female Athletes	□	V	D
I_{Safe}	Safety Index			P
v_{Inj}	Injury Rate	□	V	P
I_{Sus}	Sustainability Index			D
v_{Area}	Area of a Sport Venue (m^2)	□	V	D
v_{Cnst}	Emission of Construction (t/m^2)	□	V	D
v_{Mn}	Emission of Maintenance ($\text{t}/\text{m}^2 \cdot \text{Day}$)	□	V	D
v_{Grn}	Positive Environmental Impact ($\text{t}/\text{m}^2 \cdot \text{Day}$)	□	V	D
v_{Day}	Duration of the Olympic Games (Day)	□	C	D
v_{NetEm}	Net Emission of a Sport Venue (t)	□	V	D
I_{AuPop}	Popularity Index - Australian Model			P
I_{GlbPop}	Popularity Index - General Model			P
v_{GlbAvg}	Average Popularity - Overall	□	V	P
v_{OlAvg}	Average Popularity - Olympic	□	V	P
v_{AuAvg}	Average Popularity - Australian	□	V	P
i_{GlbPop}	Overall Popularity Subindex			P
i_{OlPop}	Olympic Popularity Subindex			P
i_{AuPop}	Australian Popularity Subindex			P
I_{Inn}	Innovation Index	■	V	D
I_{Fair}	Fair Play Index			D
v_{dope}	Proportion of Athletes Doping	□	V	D
w_{AHP}	Weight of AHP Model			
w_{Entropy}	Weight of Shannon Entropy Model			
w_{final}	Weight of the Final Model			
s_{GlbM}	Score of the General Model			
s_{AuM}	Score of the Australian Model			

Table 1: Table of Variables Used and their Descriptions

[1] □: Quantity, ■: Quality
[2] C: Constant, V: Variable
[3] D: Deterministic, P: Probabilistic

The type of variables and indices left blank are to be understood to be calculated, and therefore, varying quantities.

Table 6: Scores and ranks of 11 selected SDEs evaluated for three different host nations

	Tokyo 2020		Paris 2024		Los Angeles 2028	
SDE	**Score**	**Rank**	**Score**	**Rank**	**Score**	**Rank**
Athletics	0.8095	1	0.9685	1	0.9521	1
Gymnastics	0.7197	2	0.693	2	0.6438	2
Swimming	0.6269	3	0.5587	3	0.5701	3
Karate	0.5607	4	0.3369	10	0.2621	11
Baseball/Softball	0.5276	5	0.3527	9	0.4845	4
Skateboarding	0.5265	6	0.4175	7	0.4589	6
Sport Climbing	0.5089	7	0.4294	5	0.3946	7
Breaking	0.46	8	0.4756	4	0.3052	10
Lacrosse	0.4517	9	0.424	6	0.4751	5
Coastal Rowing	0.3248	10	0.3744	8	0.3902	8
Flag Football	0.2528	11	0.1787	11	0.3278	9

4.4 Individual Significance of Each Factor

Table 2: *Significance of Each Factor*

Factor	P-Value (Error)	Correlation	Used
TAG	0.00130	Significant	✓
YTT	0.00330	Significant	✓
VDG	0.00054	Very Significant	✓
GPD	0.00049	Very Significant	✓
GIR	0.11700	Insignificant	×
GCT	0.00230	Significant	✓
PTC	0.21000	Insignificant	×
MBC	0.00012	Very Significant	✓
PAR	0.00400	Significant	✓
DOP	0.00042	Very Significant	✓
INJ	0.06800	Moderately Significant	✓

The above table shows the corresponding P-value (or amount of error) obtained when logistic regression is applied individually one by one to each factor. A random training set consisting of 80% of the total data was compiled to determine the coefficients and the remaining set was used to test the regression and find the error. We find out that in general, the factors GIR (Gender Income Ratio) and PTC (number of Participating Countries) show no correlation with whether or not the sport is included in the Olympics. Therefore, these two factors were not considered in our final model.

5.2.2 Pickleball

Pickleball is another potential candidate to be added to the Olympics. Scoring a final score of 0.459, this sport has also surpassed the threshold to be added into the Olympics.

Another strong argument for the inclusion of Pickleball in the Olympics is its fast growth. Experiencing a surge in popularity over the past few years according to Google Trends™[10], Pickleball would have the potential to bring in both younger audiences, as well as retain an older fanbase.

Furthermore, the international governing body of Pickleball, the International Pickleball Federation (IPF), was only founded 8 years ago but already had 77 member countries. Though not officially recognized by the IOC yet, we believe that the current patterns of growth as well as the addition of Squash - another popular racket sport, indicate that Pickleball has strong potential to be added to *Brisbane 2032*, or other Olympic games further down the line.

7.2 Weaknesses

1. Our popularity index is determined by the number of searches of an SDE conducted in English using Google™. However, we did not take into account the variety of dialects in the same language, where different terms are used to address the same SDEs, as well as different search engines being used whose data are not archived in Google Trends, leading to less data collected. This can be improved by including multiple languages in the API and including data from other search engines.

2. Although combined with the Shannon Entropy Method, the AHP models still contain biases nonetheless. If another group attempts to recreate this model, it could potentially yield a different result or even result in failure.

3.2. Weight Based on Public Opinion

3.2.1. Social Media Data Scraping

Throughout the last decade, social media has become an important method for the dissemination of public opinion. Social media is important because it allows people to voice their opinions directly, without distortions or censorship. This immediacy gives users the power to address current events and social issues in real time, creating a dynamic space for open discussions.

We adopted discussions regarding SDEs in the Olympics from Reddit and Twitter, global social media platforms chosen for the diverse demographics and range of views they represent, and

their text-based nature. Specifically, Reddit represents a more liberal subset of the demographic, while Twitter represents a more conservative subset of the demographic. Adopting social media data to analyze public opinion on factors determining the inclusion of new sports in the Olympics is highly valuable. Social media platforms provide a vast, real-time repository of diverse opinions from global audiences, offering insights into trends, preferences, and concerns. By analyzing this data, decision-makers can identify public interest, engagement levels, and cultural relevance of potential sports. This approach ensures that decisions align with audience preferences, enhancing inclusivity and relevance in the Olympics.

We retrieved the most relevant comments with the following search key phrases, which are designed to reveal the most information regarding the factors that we have chosen. In the process, these search keywords were revised based on the amount and representativeness of data available for the keywords.

- "Olympics New Sports"
- "Olympics Sustainability"
- "Olympics Gender Equality"
- "Olympics Costs"
- "Olympics Inclusivity"
- "Olympics Venue Reuse"
- "Olympics Culture"

After collecting the data, we performed a keyword filter to find the more relevant comments. We continued by performing information and text analysis, as detailed below.

3.2.2. Data Analysis and Weighing

In order to discover the variation of discussion levels of the factors over multiple years, we segregated our comment data by year. For each year of data, we performed a series of processing steps:

- **Clean Up.** We employed regular expressions to remove unnecessary punctuation (such as question marks and exclamation marks), special characters (such as hashtag, at symbol, and the ampersand), Markdown tags, HTML tags, and other irrelevant elements from the comments.
- **Stop Words.** Next, we removed the "stop words" from the comments to ensure that we focus on the main content only during our analysis. "Stop words" are words that are frequently used but does not provide substantial information for our analysis. These words include "the," "of," and "in." We employed predefined stop word lists from NLTK, a toolkit for natural language processing.
- **Relevance Analysis.** We employed a pre-trained BERT model to identify the relevance between each comment and each factor. BERT, or Bidirectional Encoder Representations

from Transformers, is a transformer-based model known for its ability to capture the contextual meaning of words in a sentence. For each comment and factor pair, we can generate embeddings for both the comment and the factor using BERT.

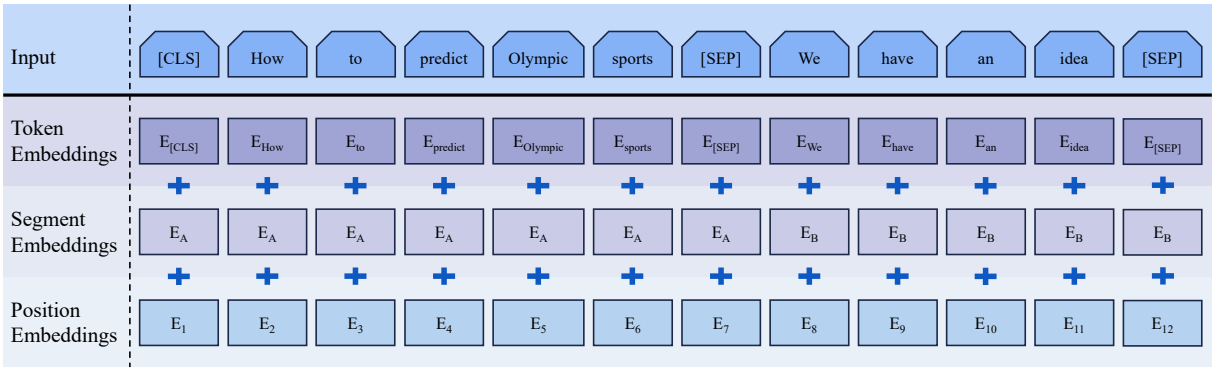


Figure 2: BERT Tokenizer

These embeddings are vectorial representations that encode the semantic meaning of the text. We identified the similarity scores by calculating the cosine similarity (dot product) between the word embeddings and our factors.

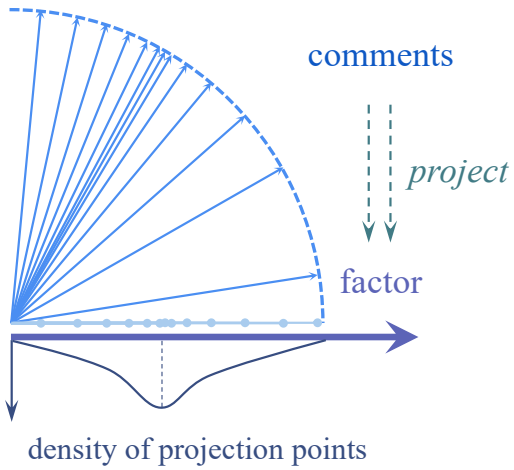


Figure 3: Cosine Similarity

Clearly, a higher similarity score indicates stronger relevance.

- **Aggregate Relevance Scores.** Afterwards, we aggregated the relevance scores for each factor across all comments in a given year. This gives us the total relevance score for each factor in each year, which reflects how much attention the specific factor received in public discussions for the year.
- **Plotting and Data Analysis.** By plotting the total relevance scores for each factor across multiple years, we can observe how public attention to various topics changes over time.

Factors with rising scores may signal growing public interest, while those with declining scores could indicate waning attention or shifting focus to other issues.

- **Decision Making.** The insights derived from this analysis can be combined with the scores from governmental data to inform decision making for the addition, removal, or reintroduction of SDEs in the Olympics.

Based on the above steps, we have obtained the following public opinion weights for our factors for the years from 2012 to 2024, where social media data are the most prevalent. For example, the following Table 2 shows the discovered weights for 2024. Weights may not add to 100% because of rounding.

Table 2: Factor Public Opinion Weights

Global Audience Size 9.09%	Streaming/TV Reach 6.35%	Equipment Cost 6.94%
Facility Availability (Global) 5.38%	Facility Availability (Local) 3.93%	Man/Woman Ratio 16%
Environmental Impact 12.33%	Venue Reusability 7.11%	Socioeconomic Diversity 8.29%
Cultural Representation 9.12%	Accessibility for Disabilities 7.26%	Safety Standards 8.18%

4.3 Modeling SDE inclusion (Problem 2)

We treat the inclusion or exclusion of an SDE as a binary classification problem. To assess the probability of inclusion of certain discipline in the Olympics, we develop a logistic regression model that incorporates three key components (Eq. 8): a base probability (P_0) that evaluates the adherence to IOC's values based on nine quantitative and qualitative subcriteria we developed, an inertia component (P_{in}) accounting for the effect of the historical inclusion or exclusion of the SDE, reflecting a "momentum effect", and a bias-correction item (P_{it}) that adjusts the models predictions to best match historical data.

$$P_{total} = P_0 + P_{in} + P_{it} \tag{8}$$

When $P_{total} > 0.5$, the discipline is included; when $P_{total} < 0.5$, it is not included. Notably, we do not build individual models for each discipline but instead create a model able to estimate any discipline, which can be achieved by training the model using historical data across all disciplines.

The logistic regression model is express as:

$$\ln \frac{P_0}{1 - P_0} = \boldsymbol{w}^T \boldsymbol{x} + b \tag{9}$$

where P_0 is the base probability of SDE being included, $\boldsymbol{x} = (X_1, X_2 ..... X_9)$ is a vector of values fro the nine subcriteria variables, $\boldsymbol{w} = (W_1, W_2 ..... W_9)$ is a vector of weights for the nine subcriteria corresponding weights, and b is the bias term. The weights w and the bias b are estimated using maximum likelihood estimation based on historical data.

The inertia component (P_{in}) is calculated as the product of a weight $\lambda_{i,j}$ and a scaling parameter η. The weight $\lambda_{i,j}$ is essentially the conditional probability of the current state (inclusion or exclusion) given the previous state. This probability is estimated from historical data (the provided HiMCM_Olympic_Data.xlsx). If the previous state was exclusion, $\lambda_{i,j}$ is set to zero. The hyperparameter η scales the effect of the inertia. The value of η can be optimized to best fit the model results to the binarized historical records.

$$P_{in} = \lambda_{i,j} * \eta \tag{10}$$

where subscripts (i, j) denotes i-th discpline at the j-th Olympic Games.

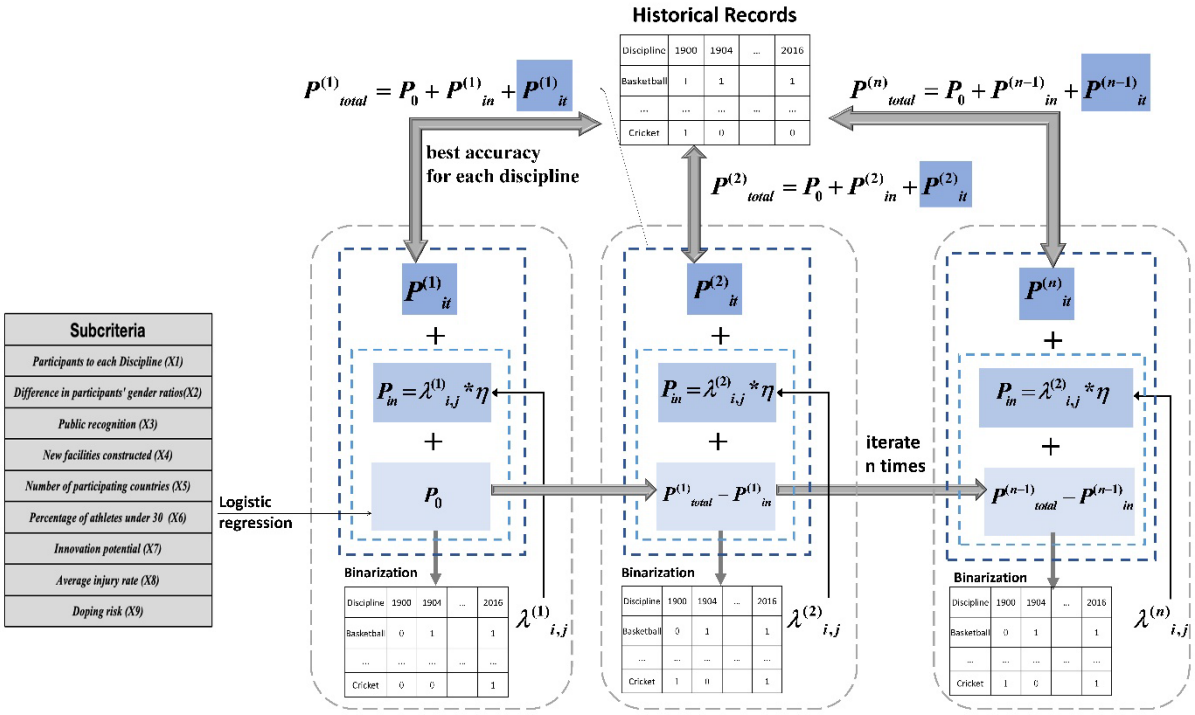


Figure 6 Diagram of the iterative modeling process

The model was implemented in Python and the iterative process is illustrated in Figure 6. The bias-correction component P_{it} is iteratively updated to improve the model's accuracy. In each iteration, the model's predictions are compared to the actual historical records, and P_{it} for each discipline is adjusted by a small increment (0.001). The direction of the adjustment (increasing or decreasing) depends on whether the model's prediction underestimates or overestimates the actual historical records.

Initially, the inertia weight $\lambda_{i,j}$ was estimated based on the classification results from the base probability P_0. In subsequent iterations, $\lambda_{i,j}^{(n)}$ was updated based on the classification results calculated from $P_{total}^{(n-1)} - P_{in}^{(n-1)}$. The iterative process continued until the model's accuracy converged to a threshold of 0.001 or a maximum of 1000 iterations was reached. To optimize the hyperparameter η, a grid search was conducted over the range of 0 to 1 with a step size of 0.1. For each value of η, the iterative process was performed, and the optimal value was selected based on the model's accuracy.

5.5 Model Comparison

The MC-TOPSIS (Monte Carlo Simulation-TOPSIS) model is introduced for comparison [7]. MC-TOPSIS is a multi-criteria decision-making method that combines Monte Carlo simulation with the TOPSIS method. TOPSIS is a system based on ideal points and negative.The sorting method of ideal points is sorted by calculating the distance between each solution and the ideal solution and the negative ideal solution. The core of MC-TOPSIS is to generate multiple possible scenarios through Monte Carlo simulation, perform multiple random perturbations on the input data to better consider the randomness or uncertainty of the input data, and finally perform statistical analysis on the ranking of decision solutions.

5.5.1 Defining Ideal and Negative Ideal Solutions

In TOPSIS, the positive ideal solution (A^+) and the negative ideal solution (A^-) are defined as follows:

1. The positive ideal solution A^+ is the vector of the optimal values for all criteria, denoted as $A^+ = \left\{y_1^+, y_2^+, \cdots, y_n^+\right\}$;

2. The negative ideal solution A^- is the vector of the optimal values for all criteria, denoted as $A^- = \left\{y_1^-, y_2^-, \cdots, y_n^-\right\}$.

5.5.2 Monte Carlo Simulation

Monte Carlo simulations were performed on the data for each assessment object to generate a range of possible scenarios to deal with the uncertainty in the data:

STEP 1: Determine the random disturbance range. Based on the data characteristics (e.g., mean and variance) of each evaluation indicator, determine the data perturbation range, which can usually be assumed to follow a normal or uniform distribution.

STEP 2: Generate simulation samples. Apply multiple random disturbances to each index value y_{ij}, generating N sets of simulation samples $\left\{Y_{ij}^1, Y_{ij}^2, \cdots, Y_{ij}^n\right\}$, where Y_{ij}^k represents the value of the i-th evaluation object on the j-th index in the k-th simulation.

STEP 3: Rebuild the evaluation matrix. The evaluation matrix generated for each simulation can be expressed as $Y^k = \left[Y_{ij}^k\right]$, where $k = 1, 2, \cdots, N$.

5.5.3 Calculate the TOPSIS Score for Each Simulation

Perform the TOPSIS calculations for each evaluation matrx Y^k generated by the sitimulation as follow procedure.

STEP 1: Standardize the Evaluation Matrix. Standardize Y^k to obatain the standardzed matrix $R^k = \left[r_{ij}^k\right]$, where

$$r_{ij}^k = \frac{Y_{ij}^k}{\sqrt{\Sigma_{i=1}^m \left(Y_{ij}^k\right)^2}} \tag{27}$$

STEP 2: Weoighted Standardize Matrix. To obtain the weighted matrix $V^k = \left[v_{ij}^k\right]$, multiplying the weight W to the standardized matrix R^k, where

$$v_{ij}^k = w_j \cdot r_{ij}^k \tag{28}$$

STEP 3: Calculate the Distances to the Positive and Negative ideal solutions.

1. The positive and negative ideal solation, A^+ and A^- respectively, are defined as

$$\begin{aligned} A^+ &= \left\{\max\left(v_{1j}^k\right), \max\left(v_{2j}^k\right), \cdots, \max\left(v_{nj}^k\right)\right\} \\ A^- &= \left\{\min\left(v_{1j}^k\right), \min\left(v_{2j}^k\right), \cdots, \min\left(v_{nj}^k\right)\right\} \end{aligned} \tag{29}$$

2. Calculate the Euclidean distance of each evaluation object to the ideal and negative ideal solutions; those are

$$\begin{aligned} D_i^{+k} &= \sqrt{\sum_{j=1}^n \left(v_{ij}^k - y_j^+\right)^2} \\ D_i^{-k} &= \sqrt{\sum_{j=1}^n \left(v_{ij}^k - y_j^-\right)^2} \end{aligned} \tag{30}$$

STEP 4: Calculate the relative closeness C_i^k. The relative closeness C_i^k of the i-th evaluation object in the k-th simulation is given by

$$C_i^k = \frac{D_i^{-k}}{D_i^{+k} + D_i^{-k}} \tag{31}$$

5.5.4 Statistical Analysis of the Results from Multiple Simulations

Repeat the above steps for N simulations to obtain the relative proximity C_i^k, of each evaluation object in each simulation, and then statistically analyze the relative proximity of each evaluation object to calculate its mean and standard deviation to obtain more stable results. The average closeness for each evaluation object can be calculated by

$$\bar{C}_i = \frac{1}{N} \sum_{k=1}^{N} C_i^k \tag{32}$$

and the standard deviantion of the closeness for each evaluation object σ_{C_i} is calculated by

$$\sigma_{\bar{C}_i} = \sqrt{\frac{1}{N-1} \sum_{k=1}^{N} \left(C_i^k - \bar{C}_i\right)^2} \tag{33}$$

5.5.5 Ranking and Result Analysis

All evaluation objects are ranked according to the average proximity $\bar{C}_i$. The higher the proximity, the higher the ranking. The standard deviation $\sigma_{\bar{C}_i}$ of the proximity can be used as a measure of the stability of the ranking results; the smaller the standard deviation, the more stable the ranking. The results can be seen. Upon comparison, it is found that the results are similar. However, the MC-TOPSIS method has a larger error band, indicating that the CWM model yields better results:

Table 7: MC-Topsis Ranking for Sports

Ranking	Event	Score	Ranking	Event	Score
1	Football	98.936430	25	Golf	47.953314
2	Athletics	96.861979	26	Canoeing	47.861390
3	Basketball	89.616694	27	Fencing	47.731547
4	Cycling	85.993677	28	Skating	47.550199
5	Volleyball	85.768341	29	Sailing	47.307150
6	Tennis	73.695559	30	Judo	47.976280
7	Gymnastics	72.723272	31	Archery	46.172152
8	Aquatics	69.875080	32	Wrestling	43.333822
9	Badminton	66.882912	33	Modern Pentathlon	42.866223
10	Table Tennis	63.737268	34	Weightlifting	40.095087
11	Ice Hockey	59.895655	35	Karate	39.672697
12	Rugby	59.366788	36	Cricket	38.979617
13	Breaking	59.318223	37	Shooting	38.923704
14	Skateboarding	58.460085	38	Water Motorsports	34.248487
15	Triathlon	58.063067	39	Lacrosse	33.593957
16	Sport Climbing	57.730372	40	Boxing	32.883900
17	Rowing	55.683576	41	Flag Football	32.769615
18	Baseball and Softball	55.434577	42	Polo	23.117768
19	Handbal	54.791165	43	Basque Pelota	19.683116
20	surfing	54.320044	44	Tug of War	18.811518
21	Field Hockey	53.189406	45	Jeu de Paume	6.883763
22	Equestrian	52.737058	46	Rackets	4.919216
23	Squash	49.971049	47	Roque	2.369794
24	Taekwondo	49.007374	48	Croquet	1.726015

6. Requirement 4 : Expand the MFD-HPC Model and Conduct Further Analysis

6.1. Comprehensive exploration of renewable energy

6.1.1. *Quantitative analysis*

To simulate the impact of increasing the proportion of renewable energy on global HPC carbon emissions, we need to plot the function of global HPC carbon emissions (C) with respect to the proportion of renewable energy (ρ). We assign ρ values of 40%, 60%, and 80% to observe the rate of change in the function. When the proportion of renewable energy is relatively low, meaning fossil fuels still dominate the energy mix, global HPC carbon emissions remain relatively stable around a value of 1 despite changes in ρ. To emphasize the impact of increasing renewable energy proportions, we applied a simple adjustment to the graph, as shown in Fig.15.

From Fig.15, it can be observed that by 2035, as the proportion of renewable energy (ρ) increases from its original value of 26% to 40%, 60%, and 80%, the total global HPC carbon emissions undergo corresponding changes. As ρ rises, carbon emissions decrease relatively; however, the overall trend remains upward, driven primarily by the growth of HPC.

Incorporating our proposed MFD-HPC Model, we have derived the following information:

Initial value : $\rho(2010) = \frac{0.9192 \cdot (2010-2010)+26.45}{100}$ = 0.2645.

Annual growth rate : $\triangle\rho = 0.9192$

When $t = 2030$, $\rho(2030) = \frac{0.9192(2030-2010)+26.45}{100}$

The proportion of renewable energy will reach 100% in year T : $\rho(T) = 0, T = 2090$

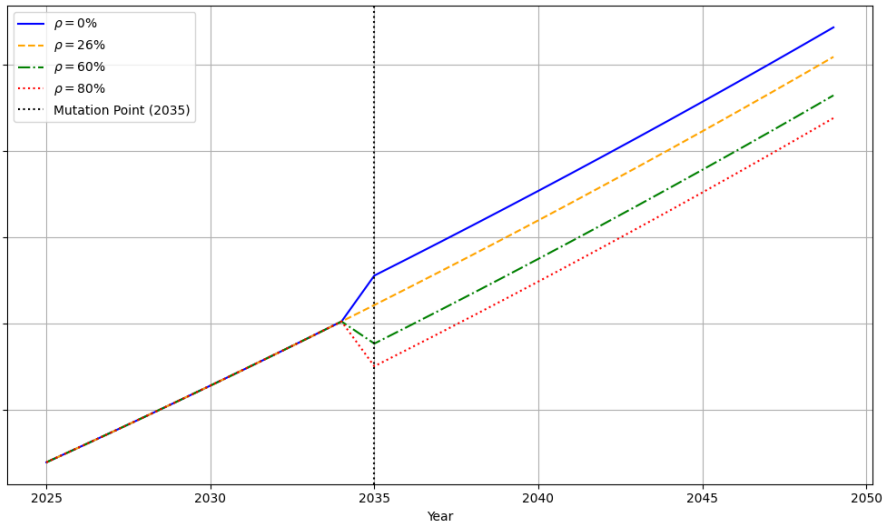


Figure 15: C_t Curves for Different ρ Values (2025-2050).

Furthermore, when the proportion of renewable energy (ρ) reaches 100% and reliance on fossil fuels and nuclear energy is eliminated, competition from other industries for energy can be disregarded due to energy saturation. In this case, global HPC carbon emissions will depend solely on the growth of HPC. As a result, the previously defined functional relationship between global HPC carbon emissions and time, expressed in Equation (5), will be simplified to the following Equation (12):

$$C_t = m \cdot (1 + r)^{t-2010} + s \tag{12}$$

In conclusion, from a quantitative analysis perspective, the proportion of renewable energy has increased from 26.45% in 2010 at an annual growth rate of 0.9192%, projected to reach 44.83% by 2030 and 100% by 2090. During this process, carbon emissions associated with fossil fuels gradually decrease and will be entirely eliminated by 2090. By 2035, when the proportion of renewable energy is approximately 76.43%, global HPC carbon emissions are primarily driven by the exponential growth of HPC, with the terms in the carbon emission formula related to fossil fuel proportions approaching zero.

When the proportion of renewable energy reaches 100%, the carbon emission formula simplifies to: $C_t = m \cdot (1+r)^{t-2010} + s$. Where carbon emissions are solely controlled by the HPC growth rate ($r = 0.4946$). Although emissions continue to rise, the growth rate decreases significantly. This indicates that increasing the proportion of renewable energy has a clear mitigating effect on carbon emissions, but the growing demand for HPC remains the primary driver of future emissions.

6.4.2 Scenario 2

HPC growth + increased share of renewable energy HPC growth maintained at 10 per cent per annum while promoting energy transition. The share of renewable energy has increased significantly to 40%, the share of fossil fuels has decreased to 45%, and the share of nuclear energy has been maintained at 15%. Clean energy supply growth is supported by policy and technology advances. The results of the simulation are shown in the figure 9 below.

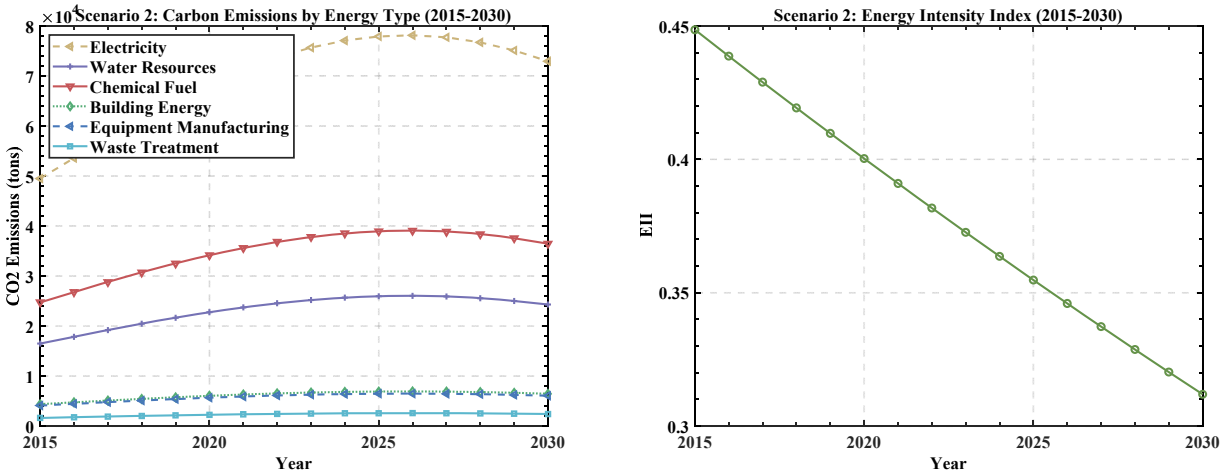


Figure. 9 Scenario 2 Simulation Results

The graph above shows that increased renewable energy supply reduces the impact of HPC growth on carbon emissions, but still does not fully neutralize the environmental burden of growth. In this scenario, the annual growth rate of carbon emissions decreases, but overall emissions may still be high, but the EII index decreases.

Boundary: Renewable energy enhancement can reduce HPC carbon emissions in 2030 to the range of million tons of CO2, emission reduction effect is obvious, but the growth rate to maintain a certain range.

3.3.3 Energy Forecast Model

We employ the ARIMA model to forecast the electricity generation of various energy sources in the coming years. This is because 1) electricity generation often exhibits significant seasonality and trends, such as wind and solar power being influenced by seasonal and weather conditions; 2) electricity generation data are often non-stationary, meaning their statistical properties (like mean and variance) change over time. The ARIMA model combines three methods: Autoregression (AR), Integration (I), and Moving Average (MA), making it particularly suitable for processing and forecasting non-stationary time series data. Below is a detailed explanation of the mathematical principles of the ARIMA model [14]:

- **Autoregression (AR):** The autoregressive part focuses on the linear relationship between the current value and past values in a time series. If the current value Y_t of a time series can be expressed as a linear combination of its past p values plus a random error term ϵ_t, then the time series has autoregressive properties.

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \ldots + \phi_p Y_{t-p} + \epsilon_t \tag{16}$$

  where $\phi_1, \phi_2, \ldots, \phi_p$ are the autoregressive coefficients.

- **Integration (I):** The integration part is the process of transforming a non-stationary time series into a stationary one. The number of differences d indicates how many times differencing is needed to make the series stationary. First differencing $\Delta Y_t = Y_t - Y_{t-1}$, second differencing $\Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1}$, and so on.

- **Moving Average (MA):** The moving average part focuses on the linear relationship between the current value and past error terms in a time series. If the current value Y_t of a time series can be expressed as a linear combination of the current and past q error terms, then the time series has moving average properties. The mathematical expression is:

$$Y_t = \mu + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \ldots + \theta_q \epsilon_{t-q} + \epsilon_t \tag{17}$$

  where μ is the mean of the series, and $\theta_1, \theta_2, \ldots, \theta_q$ are the moving average coefficients.

The ARIMA model integrates the above three components, and its general form can be expressed as:

$$\phi(B)(1-B)^d Y_t = \theta(B)\epsilon_t \tag{18}$$

where $\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \ldots - \phi_p B^p$ is the autoregressive polynomial, $\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \ldots + \theta_q B^q$ is the moving average polynomial, and B is the backshift operator, such that $BY_t = Y_{t-1}$.

2.3 Temperature Model

To calculate the annual average temperature at data center locations, we assume a linear relationship between latitude and temperature. As latitude decreases, the average annual temperature generally decreases, driven by the Earth's axial tilt and the distribution of solar radiation. This assumption aligns with statistical findings from prior research [8].

In a study involving 2,086 weather stations, researchers derived a statistically significant regression model that explains temperature variations across different locations. Notably, latitude and longitude were identified as the two most significant predictors of temperature, rendering other variables negligible for this analysis. Thus, we simplify our model by including only latitude and longitude as inputs.

Based on these findings, we define the local average annual temperature as a function of longitude and latitude:

$$T(L_1, L_2) = 39.2 - 0.56|L_1| - 0.04L_2 \tag{2}$$

The visualization of temperature as a function of longitude and latitude is presented below:

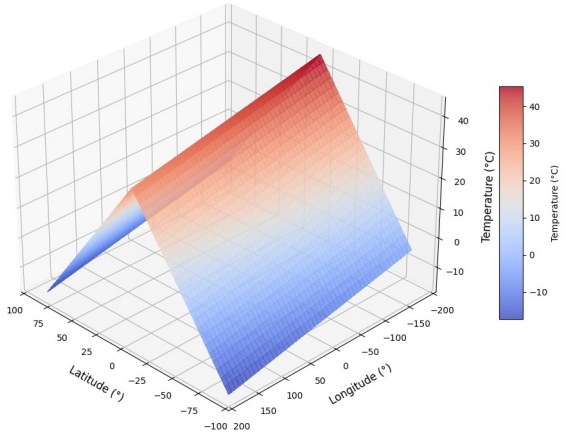


Figure 2: 3D Visualization of Temperature Based on Latitude and Longitude

4.4.3 Scenario 2: Clean Energy Transition

This scenario simulates the changes in carbon emissions under conditions where the energy structure rapidly transitions to clean energy while phasing out coal power. It aims to evaluate the isolated impact of a large-scale clean energy transition. The purpose is to reveal the effectiveness of clean energy development in mitigating carbon emissions within the current technological and policy framework, quantify its role in emission reduction, and provide insights for optimizing energy structure adjustment strategies.

The procedure is identical to that in Section 4.4.2, and five charts are similarly created.

Parameter settings:

- **HPC demand and energy efficiency predictive factors r_{HPC} and r_{eff}: Current.**
- **Annual growth rate of power generation r_{growth}: Positive.**
- **Reduction rate of transmission losses r_{TLL}: Current.**
- **Growth rate of emission reduction factors r_{ERF_gro}: Current.**

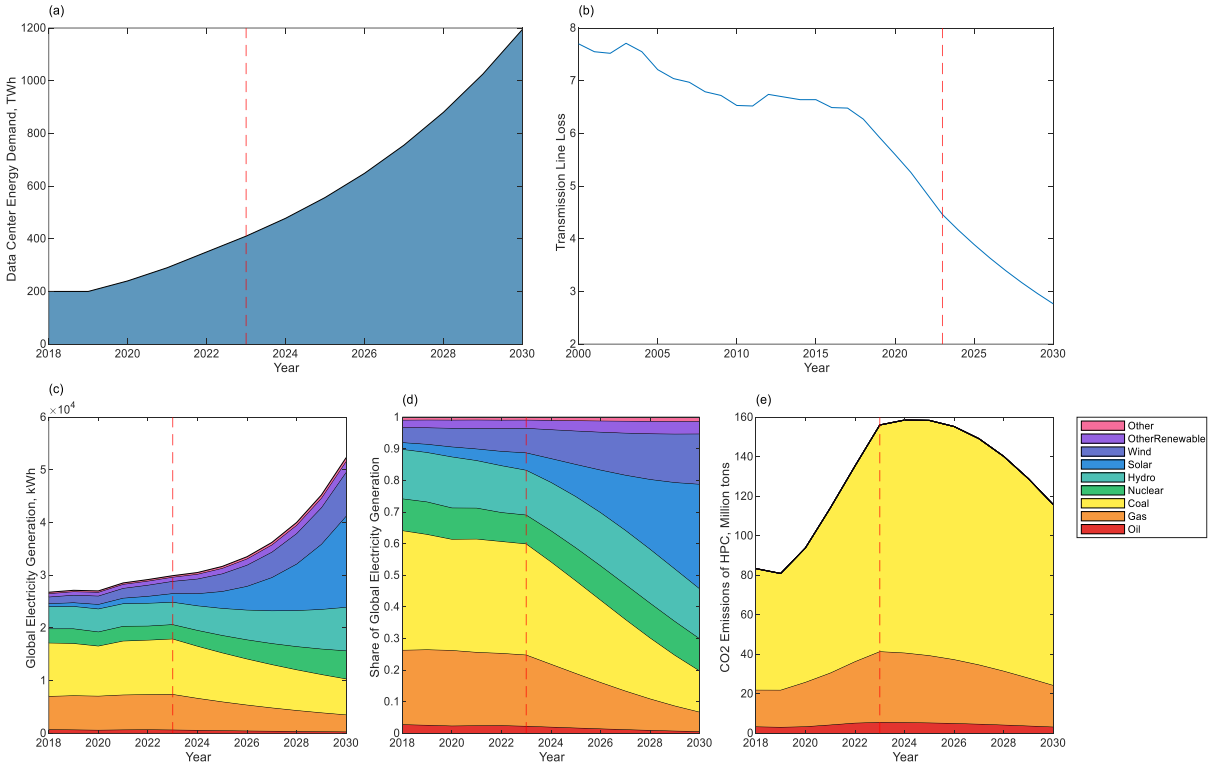


Figure 8: Clean Energy transition: (a) Projected global HPC energy demand over time; (b) Plot of transmission line loss TLL over time; (c) Chart of global energy production over time; (d) Chart of the global share of energy production over time; (e) Projection of total carbon emissions over time.

Under the assumption of accelerated clean energy growth and a gradual reduction in coal power usage, the projected changes in future energy structure and carbon emissions are analysed through a series of charts. As shown in Figure 8, the share of clean energy (such as solar, wind, and hydropower) in global power generation increases significantly, with solar and wind energy showing the most notable growth. Meanwhile, the proportion of traditional fossil fuels, such as coal and natural gas, steadily declines. This trend demonstrates the direct and significant impact of accelerating clean energy growth on optimizing the energy structure.

Further analysis reveals that, in this scenario, although global HPC demand continues to grow, the increase in the share of clean energy leads to a significant reduction in carbon emissions from HPC power generation. The decrease in coal power usage has a marked effect on reducing total carbon

emissions. Compared to the "status quo" scenario, the carbon emission curve shows a much slower upward trend, with some periods even exhibiting stabilization or a decline. This indicates that by accelerating the deployment of clean energy and phasing out high-carbon energy sources, it is possible to meet growing energy demands while significantly reducing global carbon emissions.

Additionally, the charts reveal that this scenario alleviates pressure on the transmission system, as the distributed nature of clean energy may reduce the need for long-distance power transmission. Overall, the analysis of this scenario highlights that accelerating clean energy growth and reducing the share of coal power is a key pathway for achieving a low-carbon energy transition and can play a positive role in mitigating climate change.

8 SENSITIVITY ANALYSIS

8.1 Model Robustness Evaluation

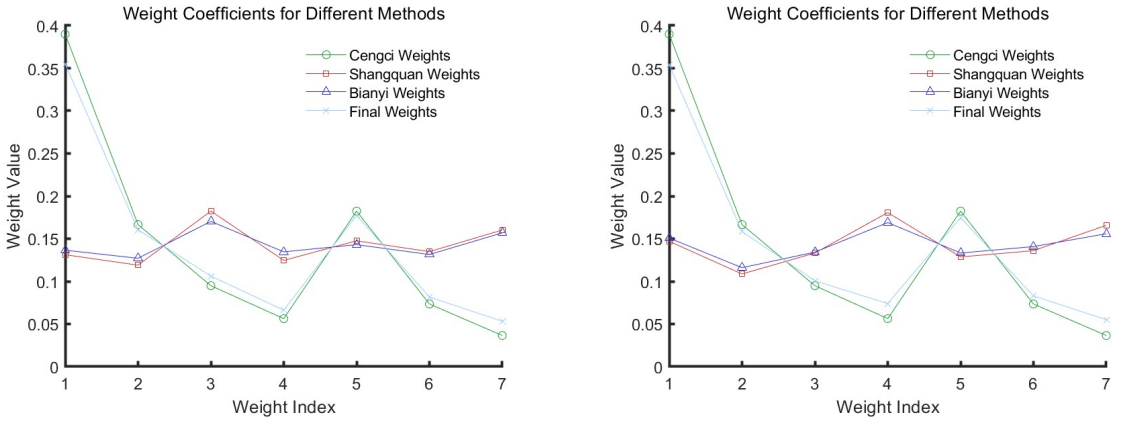
In this model, we have adopted a multi-criteria decision-making framework, incorporating comprehensive weighting methods such as the Analytic Hierarchy Process (AHP), Entropy Weight Method, Coefficient of Variation Method, and CRITIC method, along with the MC-TOPSIS model. These methods collectively construct a multi-dimensional and highly comprehensive scoring system, effectively balancing objectivity and flexibility.

8.1.1 Robustness Performance

Multi-Indicator Coverage. The model covers multiple dimensions such as popularity, gender equality, sustainability, inclusivity, innovation, fairness, and economic benefits, capable of reflecting the comprehensive value of SDEs from various aspects. The weights of multiple indicators are derived through multiple weighting methods, reducing the risk of bias.

Uncertainty Handling. MC-TOPSIS introduces Monte Carlo simulation to simulate multiple scenarios,taking into account fluctuations in different weights and changes in scores. This can reduce the extreme impact of a single score and make the final result more stable, demonstrates higher sensitivity.

Comparison of Indicator Weight. We found that the weight values of indicators 1, 4, and 7 vary significantly across different methods, indicating that these three types of indicators have a relatively large impact on the results.



(a) Weight Coefficients for Different Methods 1 (b) Weight Coefficients for Different Methods 2

Figure 6: Weight Coefficients for Different Methods

8.1.2 Potential Uncertainties

Subjectivity in Weight Allocation. Even with weight allocation through methods such as AHP and CRITIC, there may still be influence from experts' subjective opinions.

Data Dependence. The output results of the model are highly dependent on the accuracy and update rate of the data, especially those related to popularity and commercialization potential, which often fluctuate due to regional differences and variations in audience demographics.

Robustness Conclusion. In summary, the model demonstrates high robustness and fault tolerance, effectively adapting to variations in different SDEs data. However, in terms of specific weights and data dependence, the model may be influenced by certain subjective and regional factors. Continuous adjustment and updating of data during use are required to ensure the scientific validity of decision-making.

8.2 Analysis of the Advantages and Disadvantages of Factors with Higher Scores

In the SDEs model, factors contributing to higher scores include popularity and accessibility, inclusivity, economic benefits and commercialization potential, as well as relevance and innovation. Below is an analysis of the advantages and disadvantages of these factors.

Table 12: Advantages and Disadvantages of Factors with Higher Scores

Factors	Advantages	Disadvantages
Popularity and Accessibility	Projects with high popularity attract a large audience, which is beneficial for the global promotion of the Olympics. This indicator is easily quantifiable through data such as global audience size and event coverage, providing an intuitive reflection of the project's global influence.	Popularity varies significantly across different regions, and some projects may lack global appeal, potentially overlooking niche but culturally valuable projects.
Inclusivity	Projects with high inclusivity reflect multiculturalism and international participation, aligning with the Olympic spirit and enhancing the global representativeness of the Olympics.	Projects with high inclusivity may not necessarily have widespread global commercial value, which could affect the Olympics' revenue and sponsorship appeal.
Economic Benefit and commercial Potential	Projects with strong commercialization potential attract sponsorship, contributing to enhanced economic benefits, and increase visibility through media coverage.	Over-commercialization may diminish the cultural value of the Olympics, causing niche projects to be overlooked, and affecting the diversity and balance of the Olympics.
Relevance and Innovation	Innovative projects attract younger audiences, adding vitality and staying in line with modern trends, aligning with the trend of introducing emerging projects.	Innovative projects may face safety and adaptability issues, with some project rules still not fully developed, and global recognition may take time to build up.

8.3 Conclusion

Overall, the factors in the model that contribute to higher SDEs scores play a positive role in the diversification, inclusivity, and global influence of the Olympics. However, commercial factors such as popularity and economic benefits sometimes conflict with the goals of balance and diversity in the Olympic spirit [9]. To optimize future decision-making, it is recommended to introduce innovative projects moderately while retaining mainstream projects, in order to maintain the global appeal and diversity of the Olympics. Additionally, as the hosting locations of the Olympics and global sports trends change, dynamically adjust the weights in the model to enhance its robustness and adaptability.

9.1 Strengths and Weaknesses

Strengths

- **Multi-factorial Evaluation**: This model considers all aspects of the IOC criteria, ensuring a well-rounded evaluation of all disciplines.
- **Balanced Weighting Method**: By combining the subjective Analytical Hierarchy Process (AHP) and the objective Entropy Weight Method (EWM), the model's weighting process balances subjective judgment with data-driven insights, reducing potential biases.
- **Validation with Real IOC Decisions**: Testing the model against IOC's past decisions on recently added or removed SDEs, as well as continuously existing ones, demonstrates its alignment and reliability. This validation ensures the model is grounded in real-world decisions.
- **Robust Sensitivity Analysis**: By evaluating the impact of fluctuations in key factors like popularity, construction cost, and inclusivity, the model demonstrates its robust stability and resilience to minor data variations.

Weaknesses

- **Simplistic Assumptions**: Assumptions such as the exclusion of political, cultural, and social factors and the reliance on social media and news reports as representative of public opinion might oversimplify real-world complexities.
- **Limited Scope of Data Sources**: The model relies heavily on media reports (e.g., NBC and BBC) for popularity metrics. This could introduce regional bias and exclude significant global perspectives, particularly from underrepresented regions.
- **Reliance on IOC Criteria**: The model presumes that IOC decisions strictly adhere to stated criteria, disregarding potential external influences like lobbying by host countries or economic considerations as well as non-quantifiable aspects such as cultural significance or emotional appeal.

8. Conclusions

We developed a probability-based model to predict the inclusion of sports, disciplines, and events (SDEs) in future Olympic Games, aligning with the International Olympic Committee's (IOC) criteria. The model integrates a base probability (P0) assessing IOC value adherence, an inertia component (Pin) accounting for past inclusion, and a bias-correction item (Pit) to optimize historical data fit. Sensitivity analysis highlighted the importance of Popularity and Relevance and Innovation criteria, with predictions for the 2032 Brisbane Olympics suggesting Australian football, Zwift cycling, and netball as leading candidates.

Our model demonstrates high predictive accuracy and provides a data-driven alternative to traditional, subjective methods. By learning from historical trends, the model offers objective and flexible predictions while being adaptable to more granular levels with sufficient data. Challenges in data collection remain, but future efforts could further refine the model's scope and enhance its reliability for evaluating Olympic SDE inclusions.

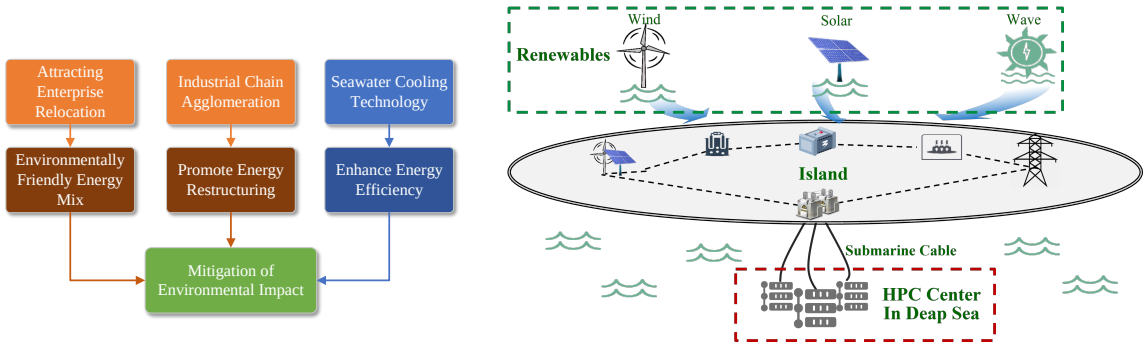
6 Recommendations of Policy and Technology

6.1 Recommendations to Reduce the Environmental Impact of HPC

We have noticed that some top technology companies are already using island new energy to power their undersea HPC centers, and at the same time using seawater as a cooling medium, setting an excellent example for improving PUE and reducing the carbon emission intensity of HPC [17, 18]. Policies and technological developments can be leveraged to guide the layout of HPC towards islands/deep-sea development, and thus we propose the following three recommendations.

1. **Attracting Enterprise Relocation Policies:** For enterprises willing to relocate their HPC centers to islands or deep-sea areas where new energy sources are more abundant, offer a discount subsidy on electricity prices based on electricity consumption.
2. **Industrial Chain Agglomeration Policies:** For enterprises willing to build new energy power generation facilities for the HPC industry in areas with abundant new energy sources such as islands or deep-sea areas, purchase electricity at a higher price to help enterprises quickly recoup investments.
3. **Developing Seawater Cooling Technology:** Provide HPC centers with better and more cost-effective cooling systems, enhancing the energy efficiency of HPC centers.

The first two policy recommendations aim to use electricity price subsidies to attract the HPC industry to relocate to island/deep-sea areas rich in new energy resources, making the energy mix of the HPC industry more environmentally friendly. At the same time, they encourage new energy companies to accelerate the construction of local new energy networks, promote energy restructuring, and reduce carbon emissions from electricity use. The third suggestion, tailored to the unique environment of islands/deep-sea areas, involves the development of distinctive seawater



(a) Recommendation Diagram (b) HPC with Energy Island

Figure 14: Impact of Recommendations

cooling technology to enhance the energy efficiency of HPC centers and further reduce the energy consumption of HPC. The predicted effects are shown as figure 14b.

Team Control Number

15926

Problem Chosen

A

2024
HiMCM
Summary Sheet

Every four years, each Olympic Games sees the introduction, removal, or reintroduction of certain sports, disciplines or events (SDEs). Our team aims to determine whether a SDE will be included in the Olympics and developed a model scoring each SDE, while the score shows the potential of SDEs being selected.

In Task 1, we conducted factor analysis based on the six IOC criteria and identified 21 factors, along with their units and characteristics.

In Task 2, we used the 21 factors to construct the evaluation model. Given our data has relatively high dimensions while no assumptions on it, we adopted **Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)** as our basic model, which aligns best with our data characteristics. To optimize model performance, we incorporated weight calculations into traditional TOPSIS method. First, we used the **entropy weight method (EWM)** to calculate the weights of each factor, and thereby obtain a score from TOPSIS integrating the weight. Next, we repeated the process using the **Analytical Hierarchy Process (AHP) method** to derive a second score. Finally, a weighted average of the two scores produces the final score, 0.6 for the first and 0.4 for the second. After the model is constructed, we confirmed its feasibility by inputting simulated data into it. Also, **we set a threshold of inclusion as 0.45,** based on the distribution of the simulation results

For Task 3, we selected three recently added SDEs (Sport Climbing, Skateboarding, and Surfing) and three long-standing SDEs (Swimming, Artistic Gymnastics, and Fencing) in the recent four Olympic Games as test subjects. After collecting data and inputting it into our model, we found new SDEs all scored above the given threshold 0.45, all **scoring high on Popularity and Accessibility principle**, especially Skateboarding; while the old SDEs were also steady above the threshold with a gradual increasing trend, **scoring similarly well on all 6 principles**. **These results align with real-world trends, demonstrating the generality of our model.**

In Task 4, we evaluated SDEs for future Olympics. Field Handball, Teqball, Rugby Union, and League of Legends were for 2032. By inputting estimated data into the model, we concluded that scores for Field Handball, Teqball, and League of Legends all indicated potential inclusion in 2032. The ranking was Teqball > League of Legends > Field Handball, with **Teqball scoring significantly high on Relevance and Innovation**. Next, we identified Baseball and Squash as potential candidates for the 2036 Olympics, performing similar process to compute their scores. **The model yielded scores above the threshold for both, suggesting they both have the potential of being selected.**

In Task 5, we conducted sensitivity analyses on both single and multiple factors. First, we performed single-variable analysis on the 21 factors and plotted the results, concluding that ***Continental Spread***, ***Injury Probability***, **and** ***F/M Athlete Ratio*** **were the most sensitive factors**. Then, we **examined intersection effects** between variables, selected two pairs of factors for multifactor analysis and plotted their results. Next, we conducted analysis for each of the six principles, **finding that** ***Inclusivity*** **was the most sensitive among them.** Lastly, since we used a weighted average method to calculate scores, we also performed a sensitivity analysis on the two weights used in the model.

Finally, we wrote a **letter to the IOC** presenting our model and findings.

Keywords: TOPSIS method, Entropy Weight method, Analytical Hierarchy Process method, Weighted Average method, Threshold of Inclusion, Single and Multiple Factor Sensitivity Analysis

4. Letter to the IOC

HIMCM Team #15926
Nov 19th, 2024

To,
Thomas Bach
President of the International Olympic Committee,
Lausanne, Switzerland

Dear Mr. Bach,

First of all, thank you! Knowing the importance of choosing suitable SDEs to the success of an Olympics game, we are very honored to contribute to the evaluation process of them, and are very happy to present our evaluation model now.

As the Olympics promotes global inclusivity and has a significant influence in the sports world, we first need to carefully consider which features of the SDEs to use as factors of our model. As for this, we used a total of 21 factors, following the six principle criteria, and including as many aspects in our model as possible.

Next, we facilitated three widely-applied evaluation models and combined them into our own integrated model. We first adopted two different methods to compute the weight of the 21 factors, the Entropy Weight Method and the Analytic Hierarchy Process (AHP) method, with the former method relying more on objective data while the latter one including more subjectivity, we can obtain two sets of weights to address the mixed subjectivity and objectivity of our data. Then we passed these weights into the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) model to generate two scores for each SDE, and finally calculate one final score using weighted average of the two scores.

After finishing model constructing and testing, we applied it to predict most possible SDEs that will be added in the 2032 and 2036 Olympics. From the outcomes of our model, we concluded that Teqball and League of Legends are most possible to be introduced in the 2032 Olympics, while field handball is likely to be reintroduced. For 2036, baseball is likely to be reintroduced.

It will be a great honor if you considered our model and suggestions. With these SDEs, we hope to maximize the six criteria given by IOC and make the Olympics as successful as possible.

Sincerely,
HIMCM Team #15926

6. Letter to the United Nations

Dear Members of the United Nations Advisory Committee on Artificial Intelligence,

On behalf of individuals and organizations deeply concerned about the development of artificial intelligence (AI) and its environmental impact, I am writing to you to express our gratitude for the insightful report "Governing Artificial Intelligence for Humanity", published by your committee in September 2024. However, I noticed that the report did not adequately address the potential environmental impacts of high-performance computing (HPC) in achieving the 2030 Sustainable Development Goals. Therefore, I urge the committee to consider prioritizing this issue in future reports and action plans. Below, I outline several reasons why this is essential:

1. Impact of Energy Consumption

HPC is a critical technology driving modern AI, machine learning, and scientific research. However, HPC clusters consume vast amounts of energy to support computational tasks, particularly for deep learning training and large-scale data processing. For instance, training a large deep neural network often requires thousands of GPUs operating in parallel, leading to immense energy demands.

According to our data, energy consumption driven by HPC has been increasing annually. While technological optimization has gradually reduced transmission losses, HPC-related energy demands continue to grow, leading to an upward trend in energy production needs from power plants. Modelling predictions suggest that this growth rate is accelerating. Research indicates that AI training's energy consumption not only depletes non-renewable resources like coal, oil, and natural gas but has also become a significant contributor to global carbon emissions, exacerbating environmental pressures.

2. Greenhouse Gas Emissions

HPC energy consumption is directly linked to emissions of greenhouse gases like CO_2. While some data centres have adopted clean energy, many still rely on traditional fossil fuels due to the disparities in global energy structures. This reliance makes AI technologies and HPC significant contributors to global greenhouse gas emissions.

Our modelling indicates that while emission reduction factors have been improving annually, reducing emissions per unit of energy generated, the overall energy demand continues to rise. Without proactive interventions, carbon emissions are projected to increase steadily in the coming years. Excessive greenhouse gas accumulation is not only the primary driver of global climate change but also triggers cascading effects that harm ecosystems, economies, societies, and human health. Addressing this issue is urgent.

3. Other Environmental Issues

Our extended evaluation models indicate that HPC poses several additional environmental challenges that merit attention:

- **Air Pollution**: In the power generation process, carbon emissions are accompanied by the release of nitrogen oxides (NO_x) and sulphur dioxide (SO_2), leading to issues such as acid rain, photochemical smog, and direct harm to human health.
- **Electronic Waste**: Rapid technological advancements have accelerated the turnover of HPC hardware, resulting in significant accumulation of electronic waste. Improper disposal of these products can have long-term negative impacts on soil, water sources, and air quality.
- **Water Consumption**: Cooling large computational facilities requires substantial water resources. Data centres in water-scarce regions face considerable environmental pressures.
- **Resource Depletion**: Many electronic components essential to HPC rely on rare metals such as aluminium, copper, cobalt, and rare earth elements. The extraction and consumption of these resources exacerbate resource depletion, environmental degradation, and socio-economic conflicts.

Recommendations

While AI and HPC have brought immense progress and innovation, their environmental challenges must not be overlooked. I propose that the United Nations Advisory Committee on Artificial Intelligence include a detailed section on the environmental impact of HPC in future reports, particularly in relation to the 2030 Sustainable Development Goals, and offer actionable policy recommendations. These could include:

1. **Investing in Green Computing Technologies**: Promoting research and adoption of energy-efficient technologies.
2. **Reducing Energy Consumption and Carbon Emissions**: Implementing stricter regulations and incentives to transition to cleaner energy sources.
3. **Optimizing Resource Management**: Enhancing resource efficiency and reducing waste generation.
4. **Improving Waste Treatment**: Developing better systems for handling and recycling electronic waste.

Such efforts will not only promote technological progress but also ensure the sustainability of the environment during the digital transformation.

Thank you for your attention to this critical issue. I look forward to seeing these suggestions adopted and contributing to a more profound global discussion and action plan regarding the environmental impact of HPC.

Sincerely,
Chen Yunzhi
Wang Yuchen
Li Peilin
Luo Xinrui

AI Use Report

The following are details of generative artificial intelligence (large language model) use in our solution.

- **OpenAI ChatGPT 4o.**

  Input: What are basic requirements for new sports to be included in the Olympics, as outlined in the Olympics Charter?

  Output: The Olympic Charter outlines several criteria for a new sport to be considered for inclusion in the Olympics. These include: 1. Governance and International Representation: • The sport must be governed by an International Federation (IF) recognized by the International Olympic Committee (IOC). • The IF must comply with the Olympic Movement's ethical standards and regulations, including anti-doping rules. 2. Popularity and Universality: • The sport must be widely practiced around the world. • For men's sports, it should be played in at least 75 countries across four continents; for women's sports, in at least 40 countries across three continents. 3. Relevance and Appeal: • The sport should have a strong youth appeal and contribute to gender equality. • It should reflect modern athletic trends and attract a global audience. 4. Compliance with Olympic Values: • The sport must uphold the Olympic spirit, including fairness, respect, and sportsmanship. • It should not compromise the health or safety of athletes. 5. Practicality for Inclusion: • The sport must fit within the existing framework of the Olympic Games, including logistical considerations like venue availability and cost management. 6. History and Development: • The sport should have a well-established history and structure. • It should have held significant international competitions over time.

  The IOC also considers flexibility in terms of the program and cultural significance, especially when choosing host-specific additions. These requirements ensure that a sport aligns with the Olympic values and logistics.

- **OpenAI ChatGPT 4o.**

  Input: How do I add BibTeX to a LaTeX project?

  Output: To add BibTeX to a LaTeX project, follow these steps:

  1. Create a .bib File

  • Create a file (e.g., references.bib) in the same directory as your main .tex file. • Add your references in BibTeX format. For example:

  @booklamport1994, author = Leslie Lamport, title = LaTeX: A Document Preparation System, publisher = Addison-Wesley, year = 1994, @articleeinstein1905, author = Albert Einstein, title = On the Electrodynamics of Moving Bodies, journal = Annalen der Physik, volume = 322, number = 10, pages = 891–921, year = 1905,

  2. Include BibTeX in Your .tex File

• Add the following commands where you want the bibliography to appear:

3. Cite References

• Use

citekey to cite a reference in the text. For example:

Einstein's work on relativity is groundbreaking

citeeinstein1905.

4. Compile the Document

• Compile your LaTeX project using the following sequence: 1. Run pdflatex (or latex) on your .tex file. 2. Run bibtex on your .aux file. 3. Run pdflatex twice more to update references and citations. Example command-line sequence:

pdflatex main.tex bibtex main pdflatex main.tex pdflatex main.tex

5. Check for Output

• Ensure the bibliography appears at the end of the document. • Verify that citations link correctly to the bibliography.

Let me know if you need help with a specific error or customization!

- **OpenAI ChatGPT 4o.**

  Input: I am trying to conduct public opinion analysis. What are some tools to do this?

  Output: For public opinion analysis, consider these tools: 1. Social Media Monitoring Platforms: • Brandwatch: Offers sentiment analysis and social media monitoring across multiple platforms. • Hootsuite Insights: Allows you to track and analyze social media mentions, hashtags, and sentiment. • Meltwater: Provides media intelligence and sentiment tracking for social media, news, and blogs. 2. Text Analysis Tools: • MonkeyLearn: A no-code text analysis tool with pre-trained models for sentiment and opinion analysis. • VADER Sentiment Analysis: Specialized for analyzing social media text, working well with Python's Natural Language Toolkit (NLTK). • IBM Watson Natural Language Understanding: Analyzes tone, sentiment, and emotion in text content. 3. Survey Tools with Sentiment Analysis: • SurveyMonkey: Collects feedback and includes built-in sentiment analysis. • Qualtrics: Supports advanced data analysis and sentiment scoring for survey responses. 4. Natural Language Processing Libraries (for custom analysis): • NLTK and spaCy: Python libraries that support custom opinion analysis models. • Google Cloud Natural Language API: Provides sentiment and opinion analysis via machine learning. 5. Reddit Analysis Tools (for specific platforms like Reddit): • Pushshift API: Collects Reddit data and can be combined with other tools (like VADER or MonkeyLearn) for sentiment analysis. • PRAW (Python Reddit API Wrapper): Useful for scraping and analyzing Reddit comments or threads.

  Combining tools can give you a more comprehensive analysis across platforms and content types.

- **OpenAI ChatGPT 4o.**

  Input: Do articles in news websites reflect public sentiment?

  Output: Articles on news websites often reflect editorial perspectives rather than direct public sentiment. Journalists may aim to report facts objectively, but articles can be influenced by the publication's editorial stance, the journalist's perspective, or an intended audience tone. Additionally, news articles generally aim to inform rather than capture the full diversity of public opinions.

  For a more accurate gauge of public sentiment, social media posts, comments, surveys, and other sources where individuals express opinions directly are often more representative. Analyzing user comments on news articles, where available, can also provide insight into public responses to news topics.

- **OpenAI ChatGPT 4o.**

  Input: Think of all terms related to diversity, sustainability, gender equality, costs, inclusivity, and culture. Put them in a Python list.

  Output: Here's a Python list containing terms related to diversity, sustainability, gender equality, costs, inclusivity, and culture:

  terms = ["diversity", "equity", "inclusion", "cultural competence", "socioeconomic diversity", "environmental sustainability", "climate impact", "carbon footprint", "renewable resources", "gender equality", "gender equity", "gender parity", "gender representation", "intersectionality", "cost-efficiency", "cost-benefit", "economic feasibility", "budgeting", "financial accessibility", "inclusivity", "accessibility", "equal access", "universal design", "inclusive practices", "cultural awareness", "cultural sensitivity", "heritage", "tradition", "cultural preservation", "multiculturalism", "social justice", "fair labor practices", "human rights", "equitable opportunities", "resource allocation", "socioeconomic status", "income equality", "racial equity", "ethnic diversity", "language access", "disability inclusion", "representation", "community engagement", "sustainable development", "ecological footprint", "environmental justice", "biodiversity", "circular economy", "carbon neutrality", "fair trade", "social responsibility", "gender inclusivity", "non-discrimination", "minority representation", "economic impact", "long-term viability", "social cohesion", "cultural diversity", "indigenous rights"]

- **OpenAI ChatGPT 4o.**

  Input: What are some regional specialty sports that have never been included in the Olympics?

  Output: Here are some regional specialty sports that have never been included in the Olympics: 1. Kabaddi (India) – A team sport involving a combination of tag and wrestling, popular in South Asia, especially India. 2. Sepak Takraw (Southeast Asia) – A sport similar to volleyball but played with a rattan ball and using feet, head, knees, and chest instead of

hands. 3. Sumo (Japan) – Traditional Japanese wrestling where competitors try to force their opponent out of a ring or onto the ground. 4. Paddle Tennis (United States) – A smaller-scale version of tennis played on a court roughly one-third the size of a regular tennis court. 5. Hurling (Ireland) – A fast-paced team sport with roots in ancient Ireland, involving a wooden stick and a small ball. 6. Lacrosse (North America) – A team sport that combines elements of field hockey and basketball, played with a netted stick to carry a small rubber ball. 7. Bandy (Russia/Scandinavia) – A winter team sport similar to ice hockey, but played with a ball instead of a puck on a larger ice rink. 8. Bocce (Italy) – A bowling game where players attempt to throw their balls closest to a target ball on a grass or clay surface. 9. Gaelic Football (Ireland) – A mix of soccer, rugby, and basketball, played with a round ball that can be kicked or punched to score. 10. Caber Toss (Scotland) – Part of the traditional Scottish Highland Games, where athletes toss a tall, heavy log (caber) end over end.

These sports have strong regional followings but have not made it to the Olympic stage.

- **OpenAI ChatGPT 4o.**

  Input: What are some regional sports in China?

  Output: China has a rich history of regional sports, many of which are deeply embedded in its culture. Some notable regional sports in China include: 1. Wushu (Traditional Chinese Martial Arts) – A comprehensive system of fighting styles, which includes forms (taolu) and sparring (sanshou), focusing on agility, strength, and precision. Though some aspects of Wushu are part of the Olympics as demonstration sports, it's not fully recognized as a competitive Olympic event. 2. Dragon Boat Racing – A water sport that involves teams paddling in long, narrow boats decorated like dragons. It is traditionally held during the Duanwu Festival (Dragon Boat Festival) and is popular in southern China. 3. Sepak Takraw (Southern China) – Although originally from Southeast Asia, the sport is also practiced in southern parts of China, especially in Guangdong, where it's known as "Rattan Ball." 4. Cuju – An ancient form of soccer originating in China, with its roots dating back over 2,000 years. It was played with a leather ball and is considered the precursor to modern football (soccer). 5. Bāduánjǐn (Eight Silken Movements) – A traditional Chinese exercise system that focuses on slow, deliberate movements for flexibility and health, often practiced in parks across China. 6. Tai Chi (Taijiquan) – A form of internal martial art known for its slow, graceful movements and breathing techniques, Tai Chi is widely practiced for health, balance, and meditation. 7. Tug of War (Chinese version) – A traditional competitive sport where two teams pull on opposite ends of a rope, often seen during local festivals and competitions. 8. Jianzi (Shuttlecock Kicking) – A traditional game where players use their feet to keep a weighted shuttlecock (jianzi) off the ground. It's a popular game in parks and playgrounds. 9. Polo – Originating in ancient China, polo was once a sport of

the nobility. It's still played in some regions, especially in Inner Mongolia. 10. Traditional Chinese Archery – Though archery is widely known, Chinese archery has its own ancient traditions, often seen in historical or cultural events.

These regional sports reflect China's diverse cultural heritage and traditional practices, some of which are still enjoyed in local communities today.

- **OpenAI ChatGPT 4o-mini.**

  Input: Give some more examples of sport that have a moderately large fanbase, but aren't included in the olympics.

  Output: Here are some examples of sports that have a moderately large fanbase but aren't included in the Olympics: 1. American Football – With a huge following in the United States and growing popularity in other regions, American football has a large fanbase but hasn't made it to the Olympics. 2. Cricket – Popular primarily in countries like India, Australia, England, and Pakistan, cricket has a massive global following, but has only appeared in the Olympics once (1900) and is not currently included. 3. Squash – Despite a large global following, particularly in countries like Egypt, Pakistan, and the UK, squash has been repeatedly denied entry into the Olympics. 4. Rugby Sevens – While rugby itself has an Olympic history (from 1900), the Sevens format, which is very popular worldwide, was only recently included (2016), and rugby's traditional 15-a-side version is still not part of the Olympics. 5. Table Tennis – Table tennis is included in the Olympics, but outside of the Games, it has a strong following, particularly in Asia, where it's a major sport. 6. Handball – While handball is in the Olympics, it's not as widely followed outside Europe. It still has a moderately large fanbase, especially in countries like France, Germany, and Denmark. 7. Bowling – Popular in many countries, especially in the United States and Asia, bowling has a sizable fanbase, but it isn't currently in the Olympics. 8. Futsal (Indoor Soccer) – A variant of soccer played indoors, futsal has a significant following, particularly in Brazil, but it is not included in the Olympics. 9. Mixed Martial Arts (MMA) – MMA has exploded in popularity globally, particularly with organizations like UFC, but remains outside of the Olympic Games. 10. Roller Hockey (Quad and Inline) – Popular in countries like Spain and Italy, roller hockey has a moderately large fanbase but isn't included in the Olympics. 11. Motor Sports (e.g., Formula 1, MotoGP) – While motor sports have a massive global following, they have not been included in the Olympic Games due to their mechanical nature and the resources involved. 12. Cheerleading – Growing significantly in popularity, especially in the U.S., cheerleading has been considered for the Olympics, but remains absent despite its fanbase.

  These sports have significant global or regional followings, but for various reasons—whether due to their nature, the expense of equipment, or governance issues—they have not been included in the Olympics.

Input: Anything else?

Output: Here are a few more examples of sports that have a moderately large fanbase but are not included in the Olympics: 1. Poker – Despite being a major competitive card game with a large global following, poker has never been included in the Olympics, though it is featured in various international competitions. 2. Surfing – Surfing has a large following, especially in countries like the United States, Australia, and Brazil. Although it is set to debut in the Olympics in 2020 (Tokyo), it wasn't previously included. 3. Ultimate Frisbee (or Disc Golf) – Ultimate Frisbee, along with its variant Disc Golf, has a large grassroots following, particularly in the U.S. and Europe, but is not currently in the Olympics. 4. Lacrosse – Popular in North America, especially in the U.S. and Canada, lacrosse has a growing global presence but is not part of the Olympics. It was once an Olympic sport (1904 and 1908), but it was removed and hasn't returned. 5. E-sports – Competitive video gaming has exploded in popularity and has a massive fanbase, but while it is recognized by the International Olympic Committee, it has not yet been included in the Olympic Games. 6. Powerlifting – Powerlifting is a strength sport with a dedicated following, especially among fitness enthusiasts. While weightlifting is in the Olympics, powerlifting, which focuses on the squat, bench press, and deadlift, is not included. 7. Speed Skating (Short Track) – Though the long-distance version of speed skating is included in the Olympics, short track speed skating is also popular globally but is often overshadowed by traditional events. 8. Trampolining – This sport, which involves acrobatic flips and jumps on a trampoline, has a following, especially among gymnastics fans, but it is often overlooked in broader discussions of Olympic sports. 9. Fell Running – A type of off-road running, typically on mountainous terrain, this sport has a strong following, particularly in the UK and Italy, but has not made it to the Olympic stage. 10. Kiteboarding – While kiteboarding has a dedicated following worldwide, it is not currently included in the Olympics. However, it was considered for inclusion in the 2024 Paris Olympics, with some forms of the sport appearing in multi-sport events. 11. Billiards/Pool – Games like snooker, pool, and carom billiards have strong followings, especially in Europe and Asia, but they have never been included in the Olympic Games. 12. Softball/Softball (Women's) – While baseball is included in the Olympics, softball (often considered its counterpart) has not been consistently included despite its strong following in countries like Japan, the U.S., and Canada.

These sports show that while they have sizable fanbases, they have yet to be recognized by the Olympics, often due to either their format, governance, or other logistical factors.

- **OpenAI ChatGPT 4o.**

  Input: What are some sports that aren't currently in the Olympics, but have the chance to be?

Output: Several sports are considered potential candidates for inclusion in the Olympics in the future. Some of the sports not yet in the Olympics but have a chance to be added include: 1. Squash: Despite strong lobbying and a dedicated following, squash has not been included yet, but its bid remains strong. 2. E-sports: The rise of competitive gaming has led to discussions about adding e-sports to the Olympics, especially due to its global popularity. 3. Cricket: Cricket was once part of the Olympics in 1900, and there is a movement to bring it back, especially given its massive following in countries like India. 4. Baseball/Softball: Although it was reintroduced for the Tokyo 2020 Olympics, it's not currently on the permanent program. There's a push for its continued inclusion. 5. Karate: It was included in the 2020 Tokyo Olympics, but its future in the Games is uncertain as it was not included in Paris 2024. 6. Roller sports (Roller skating/Inline skating): These sports are gaining popularity, especially in urban environments, and there are pushes for their inclusion. 7. Surfing: Already included in the Tokyo 2020 Olympics, but its status may vary in future Games, depending on its reception. 8. Breaking: It made its Olympic debut at Paris 2024, but its long-term inclusion will depend on its success. 9. Paddleboarding (SUP): Gaining traction worldwide, paddleboarding is under consideration due to its appeal in water sports. 10. Chess: While often contested as a sport, the growing interest in competitive chess and its global following has led some to argue for its Olympic inclusion.

These sports are often considered based on factors such as global popularity, ease of understanding, and alignment with the values of the Olympic movement.

- **OpenAI ChatGPT 4o.**

  Input: How much audience does billiard have globally?

  Output: The global audience for billiards, including its various forms such as pool, snooker, and carom, is significant but varies depending on the specific discipline and region. 1. Snooker: The most popular form of billiards in countries like the UK, China, and parts of Europe, snooker has a large following. The World Snooker Championship, one of the sport's premier events, attracts over 300 million viewers globally through television broadcasts and online streaming platforms. China, in particular, has seen rapid growth in snooker's popularity, contributing to the sport's massive fanbase. 2. Pool: Pool, especially in the United States and parts of Europe, also has a large following. Major events like the World Pool Championship draw significant viewership, and professional pool competitions often receive coverage from sports networks. The audience for pool is estimated in the tens of millions globally, especially when combined with both professional leagues and amateur players. 3. Carom (Three-Cushion Billiards): While not as mainstream as snooker or pool, carom billiards still enjoys a strong following in countries like South Korea, Turkey, and parts of Europe. Its global audience is smaller but still in the millions, especially through

specialized events like the World Three-Cushion Billiards Championship.

In general, billiards enjoys a strong and dedicated following in many regions, with millions of fans worldwide. While it may not match the global reach of sports like football or basketball, its fanbase is loyal and growing, particularly in Asia and Europe. The sport's audience is enhanced by major tournaments and media coverage, both through traditional broadcasting and streaming platforms.

- **OpenAI ChatGPT 4o.**

  Input: What's the fanbase of snooker globally?

  Output: Snooker has a substantial global fanbase, with its following growing particularly in the UK, China, and parts of Europe. While it's difficult to pinpoint an exact number, estimates suggest the following: 1. Television Audience: Major snooker events like the World Snooker Championship draw significant global viewership. For example, the 2020 World Snooker Championship reached approximately 300 million viewers, with the event broadcast across over 120 countries. 2. China: Snooker has seen explosive growth in China over the past two decades. The country is now considered one of the largest markets for the sport, with millions of fans. Events like the China Open and the Shanghai Masters have attracted growing audiences, and some snooker players from China, such as Ding Junhui, have helped boost the sport's popularity. 3. UK and Europe: In the UK, snooker has a long tradition, with events like the UK Championship, World Championship, and Masters drawing strong domestic and international audiences. The UK still represents a core fanbase, particularly with coverage from networks like the BBC. Snooker also has a solid following in countries like Ireland, Germany, and Belgium. 4. Online and Streaming Platforms: With the rise of streaming services, snooker has gained a younger and more diverse audience. Platforms like Eurosport and streaming services like DAZN, as well as YouTube highlights, have helped expand its reach. The global digital audience for snooker is in the tens of millions, especially considering the large interest in watching tournaments and highlights online.

  Overall, estimates suggest snooker has a fanbase of over 500 million globally, with significant growth in Asia, particularly China, where the sport has become a mainstream entertainment option. The sport continues to grow in popularity with younger audiences, thanks to increasing digital coverage and international events.